



Cross-boundary ambidexterity: Balancing exploration and exploitation in the fuel cell industry

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Abstract

This paper presents a new perspective on the ambidexterity debate. Basing our work on the assumption that learning is a multilevel process, we develop and empirically test whether, how, and with what consequences firms can balance explorative and exploitative technology-learning activities across their organizational boundaries. Hypotheses are tested through a longitudinal panel of data tracking the cross-boundary learning strategies of 153 companies involved with innovation in the worldwide fuel cell industry over the period 1999–2006. By juxtaposing intra- and interorganizational exploration-exploitation, we find that the pursuit of cross-boundary ambidexterity is worthwhile. Firms that engage in internal exploitation tend to balance such learning orientation with explorative interorganizational agreements. Consistently, those firms engaged in external exploitation tend to balance it with an internal focus on exploration, at least in the case of exploitative alliances involving familiar partners. Moreover, results confirm that such complementary cross-boundary strategies improve a firm's innovative performance.

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Introduction

Since the seminal contribution by March (1991), research on exploration and exploitation has been burdened with a dilemma. Although representing two different, seemingly incompatible organizational learning dynamics (Siggelkow and Levinthal, 2003), maintaining an appropriate balance between the exploration of new learning trajectories and the exploitation of old ones has been widely recognized as critical to long-term success (March, 1991; Levinthal and March, 1993). In fact, as knowledge is recognized more and more as the most strategically important resource for firms, both efficient access to and use of specialist knowledge as well as the breadth of such knowledge identify organizational learning capabilities as critical to a firm's competitive advantage (Grant, 1996). Yet, despite the increasing popularity of the ambidexterity premises introduced by Tushman and O'Reilly (1996), that is, that firms should be able to pursue exploration and exploitation simultaneously via loosely

coupled and differentiated subunits or individuals (Benner and Tushman, 2003), few empirical findings have been reported on how to accomplish this daunting challenge and the related effect on performance.

To date, research has been conducted at either the firm level or the interorganizational level. On the one hand, from an intraorganizational perspective, some authors have highlighted both the need for congruence between organizational forms and learning strategies (Brown and Eisenhardt, 1997; Siggelkow and Levinthal, 2003), and how its successful implementation through both exploration and exploitation greatly improves the quality of a firm's innovation (Katila and Ahuja, 2002) and performance (He and Wong, 2004). On the other hand, the focus of research has recently advanced beyond firm boundaries to inter-organizational learning and the challenges of balancing exploration and exploitation in decisions concerning alliance formation (Koza and Lewin, 1998; Lavie and

Rosenkopf, 2006). In fact, collaborations in innovation and R&D networks is rapidly advancing towards entirely new models of learning based on outside-oriented knowledge brokering paradigms more than on traditional, inside-oriented knowledge generation approaches (Larsson *et al.*, 1988; Lane and Lubatkin, 1998; Holmqvist, 2004). However, although acknowledging that learning is multilevel (Carley, 1999; Crossan and Berdrow, 2003), these two research streams have developed separately, neglecting the potential contributions of considering intra- and interorganizational learning as two related dynamics and their joint impact on performance.

On the basis of these premises, we investigate this gap in organizational learning research, studying how firms balance exploration and exploitation strategies across organizational boundaries and the related impact on firm innovative performance. We conclude that remaining within firm boundaries is not the only way to pursue ambidexterity; rather, firms can balance and reconcile conflicting strategies across boundaries and manage the interplay between intra- and interorganizational learning.

We empirically test our hypothesis on an unusual, comprehensive panel of data. In particular, we track the cross-boundary learning strategies of 153 companies in the fuel cell (FC) worldwide industry over the period 1999–2006.

While investigating whether firms tend to balance internal and external exploration-exploitation, we find support for our hypotheses related to cross-boundary ambidexterity. In fact, we find that firms engaged in internal exploitation balance it with processes of external new knowledge development. Consistently, those firms engaged in external exploitation tend to balance it with an internal focus on exploration, at least in the case of exploitative alliances involving familiar partners. Finally, extending previous research on the benefits of ambidexterity, result confirms that such complementary strategies across firm boundaries improve the level of a firm's innovative performance.

Theoretical background and hypotheses

Balancing exploration and exploitation through cross-boundary ambidexterity

Literatures on the determinants of organizational success in a dynamic environment tend to agree on the benefits of ambidexterity, that is, the ability of an organization to perform dichotomous activities at the same time – such as manufacturing efficiency and flexibility (Adler *et al.*, 1999), differentiation and low-cost strategic positioning (Porter, 1996), or global integration and local responsiveness (Bartlett and Ghoshal, 1989).

Although research subjects vary, these studies state that focusing simultaneously on apparently diverse types of learning is important; in other words, it is crucial to both exploit existing learning trajectories – or the use and development of things already known – and explore new, entirely different learning trajectories – or the pursuit of new knowledge, of things that might come to be known (Levinthal and March, 1993; Gupta *et al.*, 2006). Accordingly, exploitation enables organizations to engage in

refinement, implementation, efficiency, and production, while exploration implements adaptive mechanisms that require experimentation, variation, search, and innovation (March, 1991). As a result, ambidexterity involves balancing exploitation and exploration over time and across different organizational domains through alignment and efficiency in the management of current business's demand, adaptability, and responsiveness to future changes in the environment (Gibson and Birkinshaw, 2004). Therefore, balancing learning trade-offs is the only way by which organizations can counter organizational inertia and competency traps, improving both efficiency and innovativeness in their daily activities (March, 1991; Levinthal and March, 1993; Tushman and O' Reilly, 1996).

The consensus on the need for balance and its related advantages is accompanied by an even stronger agreement on the tensions and difficulties associated with such a demanding task. Exploration and exploitation reflect different organizational structures (i.e., organic vs mechanistic structures as emphasized by Burns and Stalker, 1961), different mindsets due to contrasting time horizons (March, 1991), and different organizational routines and processes (Burgelman, 2002).

The result is an underlying incompatibility between the two learning strategies: first, because firm resources are scarce by definition, that is, given a certain domain of action (e.g., a specific technological area, business unit, or organizational task), more resources devoted to exploration will necessarily imply fewer resources left over for exploration and vice versa (March, 1991; Gupta *et al.*, 2006). Second, due to their intrinsic self-reinforcing character, learning competences engender lock-ins (Arthur, 1989) or competence traps by established learning trajectories (Levinthal and March, 1988). Astonished by early successes and efficiency gains accomplished through exploitation, or challenged by the failures and promises that result from exploring new fields, organizations will maintain the same learning behaviour, that is, exploration will lead to more exploration, and exploitation to more exploitation.

In sum, exploration and exploitation involve fundamentally different logics that create tensions. However, a synergistic effect might avoid both the obsolescence of too much exploitation and the inability to appropriate the returns from new knowledge generation due to an exclusive focus on exploration. Hence, there is a need for firms to manage ambidexterity, that is, to balance the two.

Intraorganizational balancing

Despite widespread support of the balancing argument, there is considerably less agreement on how ambidexterity and the balance between exploiting existing competences and searching for new solutions can be concretely achieved. Some research focuses on internal company dynamics, stressing that ambidexterity can be achieved in different ways. Exploitation can generate clearer, earlier, and closer feedbacks than exploration, but rapid learners can progressively reduce the resources allocated to exploration in favour of gaining efficiency, so that it may be necessary to support exploration to achieve an optimal balance between the two learning strategies (Levinthal and March, 1993). The design of appropriate incentives, the role of

organizational structure, and the need to infuse a risk-taking culture within a company have been recognized as the drivers of explorative behaviour that can counter the sources of myopia induced by exploitation. Tushman and O'Reilly (1996) conceptualize the ambidextrous organization as the one able to both compete in mature markets (where costs, efficiency, and incremental innovations are critical) and develop new products and services for emerging markets (where experimentation, speed, and flexibility are paramount).

Overall, even acknowledging the difficulties of reconciling such disparate logics, these studies conclude that balancing is beneficial *per se*, without accounting for the conditions that could prevent ambidexterity and the related possibility that balance could be achieved across organizational boundaries, that is, without consideration of the external context in which organizations learn and innovate.

Interorganizational balancing

Acknowledging the increasing importance of external sources of learning and innovation, a number of studies have shifted outside organizational boundaries, investigating the ambidexterity made possible by the formation of alliances. Literature focusing on a relational view of the firm (Dyer and Singh, 1998) identifies cooperation as a strategic option that not only fosters innovative development in terms of both performance and rates (Hagedoorn and Schakenraad, 1994; Afuah, 2000; Deeds and Rothaermel, 2003), but also maximizes the utilization of complementary assets (Arora and Gambardella, 1990).

Overall, the vast amount of literature on strategic alliances offers an extremely diverse collection of collaborative forms. On the one hand, the goal of reducing transaction costs has been viewed as the principal motivation for firms to collaborate (Kogut, 1988; Pisano, 1991). Moreover, strategic alliances can also reduce economic risk, because they reduce firms' financial expenditure by enabling them to pool investments and responsibilities. This result is particularly evident in highly innovative, and therefore unstable, environments, in which uncertainty and risk of failure are extremely high (Gulati, 1995). On the other hand, the increasing concentration of alliances in R&D-intensive sectors has increased the need for a broader, resource-based perspective (Eisenhardt and Schoonhoven, 1996; Gulati, 1999; Rothaermel, 2001), as a way to either acquire or gain access to resources, knowledge, and capabilities (Mowery *et al.*, 1996; Grant and Baden-Fuller, 2004). Here, then, the theoretical background underlying the appropriateness of the exploration-exploitation framework in the interorganizational domain is evident (Koza and Lewin, 1998). Accordingly, alliances can be used to exploit complementary resources between partners, reduce risks, and promote stability. At the same time, alliances can represent a vector for transferring and absorbing partners' knowledge bases to explore new technologies and markets (Rothaermel and Deeds, 2004), as well as a mechanism to fill in the holes of firms' existing technological and geographic context, facilitated by inter-firm knowledge flows occurring through the formation of alliances and mobility of active inventors across contexts (Rosenkopf and Almeida, 2003).

Similar to the internal dynamics of exploration and exploitation discussed above, maintaining balanced interorganizational relationships has been highly valued (Lavie and Rosenkopf, 2006), with research increasingly focused on industry conditions and clusters of firms that support tendencies to either explore or exploit, while simultaneously searching for methods of balancing both tendencies.

As for the studies on intraorganizational learning dynamics, research on ambidexterity at this level assumes that organizations engage in explorative and exploitative formal agreements regardless of which learning path prevails within the organizational boundaries.

Linking intra- and interorganizational balancing

If all the studies prescribing ambidexterity were accurate, firms would be daunted by the necessity to balance conflicting logics both within and outside their organizational boundaries. Although March noted that 'finding an appropriate balance is made particularly difficult by the fact that the same issues occur at levels of a nested systems – at the individual level, the organizational level, and the social system level' (1991: 72), this two-headed challenge has been overlooked so far. In other words, it is reasonable to expect that, given the difficulties intrinsic to multiple-level, contrasting pressures to both conform and change, individual organizations might focus solely on exploration or exploitation while relegating the task of balancing the two to the nexus of relationships in which they are embedded (Gupta *et al.*, 2006).

Because the simultaneous pursuit of exploration and exploitation within a single domain involves trade-off related both to the need to allocate scarce resources (March, 1991) and to the conflicting domain-specific pressures imposed by organizational inertia and absorptive capacity (Lavie and Rosenkopf, 2006), firms can specialize in one of the two learning strategies internally while pursuing the remaining one at the interorganizational level and vice versa. To summarize, the motivation to examine the dynamics of exploitation and exploration across firm boundaries is twofold. Intraorganizational learning is the source of knowledge that can be transferred and shared between organizations engaged in collective processes of learning through networks, strategic alliances, and other forms of collaboration. At the same time, the transfer and creation of knowledge at this level contributes to intraorganizational learning. It is necessary to discover how learning occurring within organizations is related to that between organizations and how firms balance and complement learning dynamics across their boundaries. Thus:

Hypothesis 1a: Firms tend to balance exploration and exploitation across their boundaries, so that the tendency to exploit internally (externally) will be compensated by the tendency to explore externally (internally).

Hypothesis 1b: Firms tend to balance exploration and exploitation across their boundaries, so that the tendency to explore internally (externally) will be compensated by the tendency to exploit externally (internally).

In view of the need for balance between exploration and exploitation and the associated difficulties within the same domain (i.e., internally or externally), we expect firms to exploit internally while exploring externally and vice versa, given resource constraints and the need to balance conflicting pressures on either exploration or exploitation.

The performance consequences of cross-boundary ambidexterity. The fact that firms seek to balance exploration and exploitation across organizational boundaries still fails to provide evidence to support the underlying ambidexterity hypothesis. In other words, the latter, according to the original formulation advanced by March (1991), eventually should enhance firm performance. Given the conceptual differences between exploration and exploitation and their implications for strategy and structure, the performance consequences of simultaneously pursuing exploration and exploitation are receiving growing research attention.

Intraorganizational ambidexterity has been shown to benefit firms by improving their ability to develop new products (Katila and Ahuja, 2002) and financial performance (Uotila *et al.*, 2009). A direct effect of ambidexterity on firm performance was tested by He and Wong (2004) by a survey of a sample of manufacturing firms, assuming that exploration and exploitation are the two distinct dimensions of learning behaviour. Again, ambidexterity is found to increase the rate of sales growth.

More recently, the performance consequences of inter-organizational ambidexterity have been studied (Lin *et al.*, 2007), and an ambidextrous formation has been found to benefit large firms in particular, given the environmental conditions and the structure of the particular network these firms have joined.

Overall, despite supporting the ambidextrous approach because of its potential to improve company performance, results are still oriented to either intraorganizational search strategies or interorganizational learning behaviour. However, since exploration and exploitation strategies can be implemented both at the company (*intra*) and at the alliance (*inter*) levels, some redundancy is apparent in the type of knowledge generated in each exploration and exploitation category. In other words, there are reasons to believe that investments in different exploration or exploitation mechanisms across levels can lead to similar outcomes (Cockburn *et al.*, 2000). A firm may be able to acquire and access similar knowledge by refining its own knowledge base or via an exploitation alliance. If this is the case, the marginal benefit of investing simultaneously in multiple exploration- or exploitation-focused mechanisms is likely to be negative. Given such potential equifinality, at both intra- and interorganizational levels, we posit that cross-boundary exploration or exploitation strategies are substitutive, leading to a smaller firm's innovative performance. Thus:

Hypothesis 2: There is a negative contingent effect between internal exploitation (exploitation) and external exploitation (exploitation) on the performance of the innovation process.

On the contrary, and in light of the need for managing trade-offs of scarce resources between contrasting

technology-learning activities (O' Reilly and Tushman, 2004), firms will minimize redundancy in knowledge generation by specializing in either exploitation or exploration internally and pursuing the other at the interorganizational level. In other words, ambidexterity will be pursued across organizational boundaries, with internal exploitation complemented by external exploration and internal exploration complemented by external exploitation. Thus:

Hypothesis 3: There is a positive contingent effect between internal exploitation (exploitation) and external exploration (exploitation) on the performance of the innovation process.

Figure 1 summarizes the four possible permutations achieved by combining exploration and exploitation across organizational boundaries.

Of interest are the northwest/southeast and southwest/northeast diagonals. The northwest quadrant combines internal explorative learning and exploration alliances. Since both activities are targeted towards exploration, we expect them to have a negative contingent effect on a firm's innovative performance. A similar situation emerges in the southeast quadrant, which combines internal exploitative learning and exploitation alliances. Again, they are expected to have equifinality in the underlying intents, thus behaving as substitutive. To summarize, along the northwest/southeast diagonal, we expect substitutive relationships that are used to modify and develop a firm's knowledge base. However, we expect complementary relationships along the southwest/northeast diagonal. In particular, the northeast quadrant combines internal exploration with exploitation alliances, while the southwest quadrant combines internal exploitative searching with exploration alliances. The southwest/northeast diagonal represent cross-boundary ambidexterity, which we view as the ability of a firm to balance different, even contrasting, learning strategies across organizational boundaries. In other words, cross-boundary ambidexterity is here defined as the ability of a firm to take advantage of the opportunities of simultaneously balancing exploration and exploitation across its boundaries.

Research design

Research setting, sample, and data selection

A suitable empirical setting for testing our hypotheses is the emerging FC industry. FCs are defined as electrochemical

Internal exploration	Exploration overemphasized -	Inside-out cross-boundary ambidexterity +
	Outside-in cross-boundary ambidexterity +	Exploitation overemphasized -
Internal exploitation		
	Exploration alliances	Exploitation alliances

Figure 1 Cross-boundary ambidexterity and empirical predictions.

engines or devices that produce energy that combines fuel (e.g., hydrogen) and air through clean chemical reaction in contrast to conventional engines based on polluting processes like the internal combustion of fossil fuels (Hall and Kerr, 2003). This industry is at its fluid stage of development, with increasing interests motivated by its main advantages: (a) societal and environmental benefits, related to urban air pollution, CO₂ emissions, and global warming, (b) oil dependence from unstable countries, (c) technical features, in that FCs use flexible and modular technologies, which span a wide range of fuels (e.g., methane, methanol, natural gas, gasoline), although they are totally non-polluting only when they are combined with hydrogen (Pogutz *et al.*, 2009b).

Dispersed sources of knowledge, modularity, vertical disintegration, and division of labour represent the main features of this emerging industry, enhancing the propensity of firms to engage in strategic alliances both to explore new technological trajectories and to refine internally developed devices (Pilkington, 2004; Vurro and Russo, 2009). Moreover, implemented projects are risky and cost-intensive, so that careful resource allocation among technology-learning activities becomes an industry-specific zeitgeist (Pogutz *et al.*, 2009a). This becomes especially relevant when organizations compete in a highly dynamic context driven by revolutionary technological changes, in which uncertainty and risk of failure are extremely high (Gulati, 1995; Mitchell and Singh, 1996; Ahuja, 2000; Rothaermel, 2000). The potential to promote the emergence of a new industry, affecting the competitive structure of several existing sectors (i.e., automotive, utility and energy production, domestic combined heat and power plant production, mobile electronics and appliances, and battery manufacturers) along their entire value chain, makes it possible to consider FCs as a strong discontinuous technological change (Rifkin, 2002; Clark *et al.*, 2003; Hall and Kerr, 2003; Nygaard and Russo, 2008; Pogutz *et al.*, 2009b).

Moreover, the turbulence and market uncertainty that characterize the industry argue for both internal exploration, since a common dominant design has not yet emerged, and internal exploitation in the refinement of what is already known in order to outperform competitors in the race to market commercialization. As a result the balance between exploration and exploitation becomes relevant especially across organizational boundaries, given the need to combine resource scarcity with the multiple forces that foster learning at different levels.

In that our research design aimed to investigate the interplay between internal and external technology-learning activities, we tested the hypotheses on a longitudinal data set combining patenting activities and formal agreements in the global FC industry over the period 1999–2006. The time frame was selected on the basis of previous analyses of industry trends, industry-specific publications, and our own data collection, which supplied a significant amount of information about the period above.

The data collection consisted of three phases. In the first phase we identified firms engaged into intraorganizational FC-related exploration and exploitation based on their patenting activity (Katila and Ahuja, 2002). The United States Patent and Trade Office database was the primary

source of FC patenting data collection. The European Patent Office and the Japanese Patent Office were used to back up the study. In so doing, we were able to track patents pertaining exclusively to FC-related technological domain. We content analysed patent documents including technology areas such as, FC solid electrolytes, electrodes, catalysts, membranes, and systems. To facilitate the patent search, International Patent Classification (IPC) was employed¹ and compared with US Patent Classification (USPC).² Patents granted were tracked according to application date, which provided a closer link in time between the intrafirm learning processes and their recording (Hall *et al.*, 2001). Patent data collection identified an initial sample of 487 companies.³

In the second phase of the data collection, our task was to identify and obtain data on each firm's formal FC-based interorganizational agreement as the locus of external exploration-exploitation. Accordingly, data were first collected from formal worldwide collaborations completed throughout the period 1999–2006 and related to the companies that had produced patented FC-related innovations. The data for interorganizational arrangements were coded from Fuel Cell Today, an industry publication that reports information on firms and the formal, signed arrangements in which they are involved (e.g., announcement date, partner identity, agreement type). Most alliance announcements were cross validated by consultation of corporate websites, the Factiva database, and/or LexisNexis database. Via this procedure, we identified 664 FC-based formal alliances.

In the third and final phase, we matched patent data with agreement data and aggregated them at the firm level. Given our interest in testing the existence and performance consequences of cross-boundary ambidexterity, the firm-year represents the unit of analysis that captures firm-level tendencies. Hence, the data for both patenting activity and alliances were transformed to firm-year observation by pooling the data across all alliances formed by each firm in a given year. The final sample is an unbalanced panel data set including 732 firm-year observations referred to 153 unique partners.⁴ A description of the sample is provided in Table 1.

Variable definitions

Based on the goal of the study, our analysis was performed through two subsequent steps. First, was the cross-boundary ambidexterity hypotheses via internal and external exploration-exploitation, respectively, as dependent and independent variables and vice-versa. To achieve clarity, Table 2 summarizes the composition of dependent and independent variables in the models we developed to test Hypotheses 1a and 1b, as well as the predicted signs. Each variable is further described in the section on independent variables that follows.

In each model, the dependent and independent variables were lagged by 1 year (e.g., in Model 1 external exploration is referred at time $t + 1$, while the independent variables are considered at time t). We checked the reliability of such a time frame both with experts in the field and on the basis of our own reading of signed alliances that identify the technology on which the alliance was based both in the case

Table 1 Sample description

	N	%
Firm	418	
With registered FC patents	267	0.64
Size (average number of employees)		
Small firm (<50 empl.)	75	0.21
Medium-sized firm (>50 and <250 empl.)	54	0.15
Large firm (>250 empl.)	237	0.64
Turnover (average; USD/000)	310,652	
Profit before taxes (average; USD/000)	4,091	
Agreements	664	
R&D	358	0.54
New partner	491	0.74
Attribute	414	0.62
Geographic area		
North America	1,512	0.45
Asia	816	0.25
Europe	1,008	0.30
Industry		
Rubber and miscellaneous plastics products	209	0.50
Health care services	56	0.13
Railroad transportation	42	0.10
Food and kindred products	37	0.09
Other	74	0.18

of further development (exploration alliances) and in cases of licensing, commercialization, or refinement (exploitation alliances). Alliances supposedly predict patent application the year after agreements are signed, since knowledge and competence flows already exist well before an alliance is signed. Moreover, we decided to maintain this assumption on the basis of the pace of change in the industry and on the fact that perfect correspondence between a signed alliance and its output is not the main issue of our study. In fact, we are interested in mapping the cumulative typology of learning activities that firms are simultaneously performing in a given year.

In the second step of the analysis, we tested the innovative performance consequences of cross-boundary ambidexterity. Accordingly, innovative performance was the criterion of our dependent variables and internal exploration-exploitation and external exploration-exploitation were the independent variables.

Dependent variables

Innovative performance

Consistent with previous research, we measured innovative performance on the basis of a firm's patenting rate (Hagedoorn and Schakenraad, 1994; Henderson and Cockburn, 1994; Shan *et al.*, 1994; Ahuja, 2000). To specifically assess the firms' innovative performance within the FC paradigm, we measured innovative output by the

Table 2 Balancing hypothesis disentangled

Dependent variable	Hypothesis 1a			Hypothesis 1b		
	Case 1		Case 2	Case 1		Case 2
	Function external exploration Structure external exploration Attribute external exploration	Internal exploration ($t+1$)	Internal exploration ($t+1$)	Function external exploration Structure external exploration Attribute external exploration	Internal exploration ($t+1$)	Internal exploration ($t+1$)
Independent variables	Internal exploration (+)	Function external exploration (+)	Function external exploration (+)	Internal exploration (+)	Function external exploration (+)	Function external exploration (+)
Controls	Internal exploration	Structure external exploration Attribute external exploration	Structure external exploration Attribute external exploration	Internal exploration	Structure external exploration Attribute external exploration	Structure external exploration Attribute external exploration

number of *FC patents granted by application year*. Since it takes time for firms to convert internal and external learning into output (innovation performance), we assumed a decay time of 2 years (Powell *et al.*, 1996).⁵

Independent variables

Internal exploitation-exploration

According to previous research, we assessed intrafirm search activities over time through firms' patenting activity. In particular, we relied on the study by Katila and Ahuja (2002) to develop indicators for intrafirm exploration and exploitation, on the basis of the theoretical distinction between search scope and search depth. Accordingly, *internal exploitation* was measured as the accumulation of search experience with the same knowledge elements, with the assumption that refinement in the knowledge of a certain technology can depend on how often it is used over time. Thus, the internal exploitation variable was created by calculation of the number of times that, on average, each citation in year *t* was used during the past 3 years.⁶ *Internal exploration* was measured as the proportion of previously unused citations in a firm's focal year list of citations. We assessed the share of citations in a focal year as citations that could not be found in the previous 3-year list of patents and citations by that firm.

External exploration-exploitation

Interfirm technology-learning activities were first documented on the basis of formal FC-related agreements that the sampled firms had formed. On the basis of information availability and the difference in the mechanisms governing alliances with research institutions and firms, we decided to focus on the subset of agreements signed between firms. In the next step, we content-analysed each alliance description to ensure that the focal alliance indeed pertained to FCs and to divide a firm's total FC alliances into exploration and exploitation agreements. Following a well-established coding procedure in prior research (Koza and Lewin, 1998; Rothaermel, 2001; Park *et al.*, 2002; Lavie and Rosenkopf, 2006), we coded grants, research, and R&D alliances as exploration alliances (*function external exploration*), since our focus was on the basic research-oriented upstream knowledge discovery activities of the value chain. We identified manufacturing, licensing, development, and supply alliances as exploitation (*function external exploitation*), because they focus on the downstream knowledge-leveraging activities of the value chain. Two research assistants, who were blind to each other, and the theory to be tested coded the alliance data in two subsequent moments in time. The inter-rater reliability rate was 98%, well above the recommended threshold of 70% (Cohen *et al.*, 2003).

To capture the multifaceted nature of exploration and exploitation through formal agreements, we utilized additional measures of external learning activities. Based on recent advancements in the field (Lavie and Rosenkopf, 2006; Lin *et al.*, 2007), we assigned two additional measures to the usual function-based distinction characterizing the propensity of an alliance either to generate new knowledge or leverage existing knowledge. That is, we

added a measure of external exploration-exploitation based on the propensity of firms to enter into alliances with new partners. Accordingly, alliances with recurrent partners can be viewed as a form of external exploitation (*structure external exploitation*), in that they rely on already-established routines and leverage on prior experience and interfirm trust (Gulati, 1995; Gulati, 1999). Alliances with new partners can be considered as a form of external exploration (*structure external exploration*), in that they provide firms with new opportunities to broaden their knowledge base, though with greater uncertainty and risk. In other words, unknown partners are unlikely to share both languages and routines, thus slowing down the flow of knowledge and information and the efficiency of collaboration. Finally, we measured exploration and exploitation on the basis of similarity between partners in terms of the organizational characteristics associated with their participation in the same technological domain. In this context, firms' propensity to form alliances with partners in the same industry can be categorized as a form of external exploitation (*attribute external exploitation*), due to familiarity among partners that foster experiential learning and specialization. In contrast, external exploration (*attribute external exploration*) occurs when firms deviate from known patterns, experimenting with partners from different industries.

Controls

Several firm-specific factors were included in the analysis to account for potential heterogeneity in the tendency to engage in exploration and exploitation at the firm and industry level. Consistent with previous studies, we controlled for *firm size*, which has been shown to affect exploration-exploitation and the propensity to form alliances (Rothaermel and Deeds, 2004), measured as the log of the number of employees. Similarly, we controlled for *financial performance* measured as the log of firm net income at time *t* and *firm solvency* as the debt-equity ratio. Financial data were collected from a number of sources, including the Orbis database and annual financial reports. To assure that a firm's patenting in FC is not the result of an organization's overall patenting strategy, we explicitly controlled for the granting of all non-FC patents in a given year. We computed non-FC patenting propensity of firms (*non-FC patents*) as the log of the total number of non-FC patents by application date. We included a variable to encompass the extent to which firms primarily focus on the FC industry (*specialization*). We relied on the revealed technological advantage to account for specialization, measured as the ratio between the share of FC patents granted to firm *i* for a given year divided by the total share of FC patents granted to the *n* firms in the sample for a given year, and the share of patents granted to firm *i* in all patent classes for a given year divided by the total share of patents granted to the *n* firms in the sample in all patent classes for a given year. Specialization varies around 1, such that a value higher than 1 suggests that a firm is comparatively specialized in the technology in question, given its overall innovative performance. Path dependence has been widely recognized as the most relevant constraint to the development of ambidexterity both within and

outside corporate boundaries (Levinthal and March, 1993; Lavie and Rosenkopf, 2006). To account for potential heterogeneity based on experience, we included *firm age* as a control for the analysis, measured as the difference between the focal year and the year in which the firm applied for the first FC-related patent. Finally, we controlled for industry effect using two-digit US Standard Industrial Classification codes (*industry dummy variables*), for temporal effects (*year dummy variables*), as well as for country effect considering the geographical location of each firm in the sample (*country dummy variables*). Since industry, year, and country dummy variables are not statistically significant, we did not show those variables in the resulting table, thus facilitating analysis of the rest of the information.

Estimation procedure

Both the variables proxying external exploration-exploitation and innovative performance are count variables, and thus take on only non-negative integer values (e.g., the number of FC-related patents or the number of R&D agreements for a firm in a particular year). Poisson estimation provides a natural baseline model for such count data (Hausman *et al.*, 1984). A Poisson specification, however, requires that the mean and variance of the event count are equal. This restrictive assumption is unlikely to hold for pooled cross-sectional count data in the social sciences. Indeed, we conducted tests for over-dispersion on each of the dependent count variables, and found that the data violated the assumption of mean and variance equality. In such cases, the negative binomial estimation provides a significantly better fit for the data than the more restrictive Poisson model. Negative binomial regression accounts for an omitted variable bias, while simultaneously estimating heterogeneity (Hausman *et al.*, 1984; Cameron and Trivedi, 1986).

Given our data structure with multiple groups but with a limited history of data for each group, we performed a pooled negative binomial estimation regression. We used the Cook-Weisberg and the White test statistics to check for the homoskedasticity assumption and found the presence of heteroskedasticity (White, 1980; Cook and Weisberg, 1983). To correct for this, we used a robust-cluster estimator of the standard errors in the regressions. The robust-cluster variance estimator provides correct standard errors in the presence of any pattern of heteroskedasticity. It also remains valid and provides correct coverage in the presence of any pattern of correlation among errors within units. This estimator allowed us to relax the assumption of independence of errors in the regression. Since we used a pooled time-series approach, repeated observations may create correlated error terms and inflate *t*-statistics without this correction. In fact, the robust-standard errors are unaffected by the presence of unmeasured firm-specific factors that correlate errors of observations for the same firms, or, for that matter, any form of within-unit error correlation. Thus, the robust-cluster estimator produces correct standard errors even when the observations are correlated within clusters (STATA, 2005).

Progressing further, and shifting attention to the potential impact of external exploration-exploitation activities on internal exploration-exploitation strategies, we estimated the models by proxying internal exploration-exploitation and performing a pooled ordinary least square regression model, given the underlying nature of the dependent variable.

To test the performance consequences of cross-boundary ambidexterity (Hypotheses 2 and 3), we needed to interact the variables measuring internal and external exploration-exploitation. To avoid multicollinearity and enhance the interpretability of the results, we standardized all independent variables prior to both entering them into the various regression models and creating their cross products to test the interaction hypotheses. As illustrated by Table 3, all of the bivariate correlations fall below the recommended 0.70 threshold. To further assess the threat of multicollinearity, we calculated the variance inflation factors for each coefficient, which were consistently below the rule-of-thumb cut-off of 10 (Neter *et al.*, 1990). As explained in the previous section, we paid significant attention to appropriate time lags between our independent and dependent variables, in order to account for potential simultaneity bias.

Results

Descriptive statistics and correlations

Table 3 presents descriptive statistics and the bivariate correlation matrix, while Tables 4–7 present the regression results referred to the several models used in this study. Since several control variables were included in each model, a hierarchical regression procedure was implemented: the control variables were entered in the equation first, followed by the predictors. Models 1–6 present results relating the external exploration and exploitation activities to internal exploration and exploitation; on the contrary, Models 7 and 8 investigate the opposite relation, linking internal exploration and exploitation to external exploration and exploitation. Finally, Models 9–11 contain the interaction terms to access simultaneously the theoretical model in terms of the influence of the cross-boundary ambidexterity on the innovation performance.

Results of hypotheses test

Results presented in Table 4 refer to the first case suggested through Hypotheses 1a and 1b that relate, respectively, the tendency to exploit and explore internally to the tendency to explore and exploit externally. We found support for Hypothesis 1a when firms tended to balance internal exploitation strategies with external exploration strategies. In Models 1b, 3b, and 5b, the correlations between internal exploitation and function, structure, and attribute and external exploration are positive and statistically significant ($P < 0.05$). On the other hand, Hypothesis 1b is not supported, as proved through investigation of the relation between internal exploration and external exploitation; coefficients in Models 2b, 4b, and 6b are not statistically significant. Finally, there was the positive (respectively, in Models 2b, 4b, and 6b) correlation between

Table 3 Descriptive statistics and correlations

Variable	Mean	SD	Min	Max	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 Employees	7.93	3.27	0.00	13.08														
2 Net income	11.53	3.31	-24.66	21.28	0.68**													
3 Specialization	5.40	17.84	0.00	112.54	-0.09**	-0.02												
4 Age	44.59	42.51	0.00	204	0.52**	0.4**	-0.09**											
5 Solvency	0.10	6.20	-299.54	24.78	0.02	0.04	0	0.03										
6 Non-FC patents	3.07	2.29	0	8.32	0.59**	0.44**	-0.66**	0.41**	-0.01									
7 Int. Expt.	0.36	1.02	0	13.51	-0.12**	-0.1*	0.09*	-0.07	-0.01	-0.03								
8 Int. Expt.	0.81	0.33	0	1	0.08*	0.11*	-0.06	0.02	0.03	0.02	-0.63**							
9 Ext. Expt. (Function)	0.16	0.57	0	8	0.05*	0.13**	0.13**	0	-0.03	0.06	0.04	-0.05						
10 Ext. Expt. (Function)	0.13	0.52	0	8	-0.02	0.08**	0.14**	-0.03	0	-0.02	0.06	-0.09*	0.4**					
11 Ext. Expt. (Structure)	0.23	0.71	0	9	-0.01	0.11**	0.14**	-0.02	-0.02	0	0.05	-0.07	0.61**	0.64**				
12 Ext. Expt. (Structure)	0.06	0.37	0	8	0.07**	0.11**	0.12**	-0.01	0	0.05	0.04	-0.05	0.55**	0.6**	0.36**			
13 Ext. Expt. (Attribute)	0.23	0.76	0	11	0.04	0.14**	0.13**	-0.01	0	0.06	0.02	-0.02	0.46**	0.37**	0.48**	0.29**		
14 Ext. Expt. (Attribute)	0.02	0.19	0	4	0.1**	0.06*	0.06**	0.03	0.01	0.05	0.02	-0.01	0.2**	0.15**	0.16**	0.2**	0.26**	
15 Innovation performance	0.83	3.54	0	76	0.2**	0.15**	0.12**	0.06**	0	0.27**	0.08*	-0.08*	0.27**	0.19**	0.21**	0.26**	0.26**	0.21**

** $P < 0.01$, * $P < 0.05$. $n = 732$.

internal exploitation and external exploitation strategies (coefficients are strongly significant, $P < 0.01$, for structure and attribute external exploitation, while marginally significant, $P < 0.10$, for function external exploitation). These results reveal a contrasting tendency among the firms, which still balance internal exploitation activities, with external exploitation, which are substitutive in nature.

Tables 5 and 6 present our results of the hypothesized test relating external exploitation to internal exploration activities (Hypothesis 1a), and external exploration to internal exploitation activities (Hypothesis 1b). That is, Table 5 presents the results associating external exploitation strategies with internal exploration strategies. Hypothesis 1a is tentatively supported, since the correlation between attribute external exploitation and internal exploration is positive and marginally statistically significant ($P < 0.10$). Our Hypothesis 1a is not supported in the test for the relation between the function and structure external exploitation and internal exploration. Finally, we found an interesting negative (respectively, in Models 7b and 7d) and a statistically significant ($P < 0.05$) relation between the function and attribute external exploration and internal exploration, confirming the expected negative relation between the two strategies, which are substitutive in nature. On the other hand, results in Table 6 provide evidence that firms tend to balance exploration and exploitation across their boundaries, so that the tendency to explore externally is complemented by the tendency to exploit internally when function exploration alliances are implemented (in Model 8b). Moreover, Hypothesis 1b is not supported in the case of structure and attribute exploration alliances, which are not compensated by the tendency to exploit internally.

Table 7, located at the last stage of our analysis, presents the results of the hypotheses that required testing the effects of cross-boundary ambidexterity on firm innovation performance. Hypothesis 2 posits that internal exploitation and exploration represent a similar strategic intent on the part of external exploitation and exploration activities, respectively, since they are by nature substitutes. Thus, substitutive activities are expected to have a negative impact on the innovation performance of the firm. Models 9b, 10b, and 11b in Table 7 include all the interactions required to test the hypothesized effect on the innovation performance. We found statistically significant support ($P < 0.05$) for Hypothesis 2 when function external exploitation and exploration interact with substitutive internal exploitation and exploration, since the coefficients reveal a negative relation with the innovation performance. On the other hand, with Hypothesis 3 we suggest that ambidexterity across internal exploitation and exploration over external exploration and exploitation complement one another, thus having a positive effect on the innovation performance of the firm. Results in Table 7 (Model 9b) reveal a positive and statistically significant ($P < 0.05$) effect of the interaction between internal exploitation and function external exploration on innovation performance of the firm, which thus supports our Hypothesis 3. Hypothesis 3 is not supported in the case of structure and attribute external exploration, which interact with internal exploitation.

Table 4 Coefficients for dependent variable: external exploration and external exploitation

	<i>Ext. Expr. (Function)</i>				<i>Ext. Expt. (Function)</i>			
	<i>Model 1a</i>		<i>Model 1b</i>		<i>Model 2a</i>		<i>Model 2b</i>	
	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>
Constant	−8.1**	1.38	−8.12**	1.26	−5.87**	2.03	−6.19**	2.26
Employees	0.57**	0.14	0.51**	0.13	0.3*	0.14	0.27*	0.12
Net income	−0.09*	0.04	−0.06†	0.03	−0.04	0.06	−0.02	0.05
Specialization	0.02**	0.01	0.02*	0.01	0.00	0.01	0.00	0.01
Age	0.00	0.00	0.00	0	0.00	0.00	0.00	0.00
Solvency	0.15	0.09	0.96**	0.38	0.06	0.15	0.91	0.65
Non-FC patents	0.13	0.11	0.16	0.11	0.02	0.14	0.06	0.14
Int. Expt.			0.81**	0.26			0.41†	0.23
Int. Expr.			0.86	0.61			0.19	0.79
Chi square	1,128.240		929.700		1,184.790		1,005.570	
Log likelihood	−236.450		−214.200		−208.320		−183.770	
P <	0.000		0.000		0.000		0.000	

	<i>Ext. Expr. (Structure)</i>				<i>Ext. Expt. (Structure)</i>			
	<i>Model 3a</i>		<i>Model 3b</i>		<i>Model 4a</i>		<i>Model 4b</i>	
	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>
Constant	−5.98**	1.22	−6.69**	1.25	−22.31**	2.61	−21.78**	2.79
Employees	0.38**	0.11	0.33**	0.1	0.6*	0.28	0.53†	0.29
Net income	−0.07†	0.04	−0.05	0.03	−0.09	0.09	−0.05	0.08
Specialization	0.02*	0.01	0.01†	0.01	0.00	0.01	0.01	0.02
Age	0.00	0.00	0.00	0.00	−0.01†	0	−0.01*	0.01
Solvency	0.05	0.09	0.45	0.37	0.28	0.38	2.42†	1.46
Non-FC patents	0.05	0.1	0.05	0.1	0.15	0.21	0.35	0.29
Int. Expt.			0.68*	0.29			0.78**	0.28
Int. Expr.			1.26†	0.71			0.66	1.25
Chi square	1,234.590		1,370.360		1,163.680		1,108.200	
Log likelihood	−279.100		−249.661		−142.120		−126.640	
P <	0.000		0.000		0.000		0.000	

	<i>Ext. Expr. (Attribute)</i>				<i>Ext. Expt. (Attribute)</i>			
	<i>Model 5a</i>		<i>Model 5b</i>		<i>Model 6a</i>		<i>Model 6b</i>	
	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>
Constant	−8.04**	1.64	−7.81**	1.68	−21.2**	2.53	−24.15**	3.2
Employees	0.55**	0.16	0.48**	0.16	0.36†	0.21	0.46	0.31
Net income	−0.08	0.05	−0.05	0.04	−0.03	0.09	0	0.07
Specialization	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Age	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Solvency	0.15	0.13	1.18*	0.6	0.05	0.11	0.72	0.45
Non-FC patents	0.09	0.15	0.15	0.15	0.06	0.15	0.06	0.16
Int. Expt.			0.64*	0.27			1.11**	0.45
Int. Expr.			0.70	0.67			1.28	1.28
Chi square	672.680		640.670				2,077.180	
Log likelihood	−250.415		−226.178		−160.320		−141.350	
P <	0.000		0.000				0.000	

** $P < 0.01$, * $P < 0.05$, † $P < 0.10$.

Discussion and conclusion

Summary

The purpose of this study was to empirically test whether, how, and with what performance consequences, firms can explore new technological domains while exploiting their knowledge base across their organizational boundaries.

Consistent with Hypothesis 1a, we found that firms are complementing the internal exploitation of existing technological trajectories with more explorative agreements, in terms of alliance content, familiarity with and organizational characteristics of the partners involved. Additionally, the test of the reverse causality Hypothesis 1a is still supported, with external exploitation fostering internal

Table 5 Coefficient for the dependent variable: internal exploration

	<i>Model 7a</i>		<i>Model 7b</i>		<i>Model 7c</i>		<i>Model 7d</i>	
	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>
Constant	0.73**	0.19	0.68**	0.15	0.7**	0.15	0.68**	0.14
Employees	0.04**	0.01	0.04**	0.01	0.04**	0.01	0.04**	0.01
Net income	0	0.00	0	0.00	0	0.00	0	0.00
Specialization	0	0.00	0	0.00	0	0.00	0	0.00
Age	0 [†]	0.00	0*	0.00	0*	0.00	0*	0.00
Solvency	0	0.06	0	0.06	0.01	0.06	0.01	0.06
Non-FC patents	−0.01	0.02	−0.01	0.02	−0.01	0.02	−0.01	0.02
Ext. Expr. (Function)			−0.03*	0.02				
Ext. Expt. (Function)			0.03	0.03				
Ext. Expr. (Structure)					0.00	0.03		
Ext. Expt. (Structure)					−0.02	0.04		
Ext. Expr. (Attribute)							−0.05*	0.02
Ext. Expt. (Attribute)							0.04 [†]	0.03
R ²	0.400		0.416		0.409		0.417	
F	5.290		5.740		5.010		6.020	
P <	0.000		0.000		0.000		0.000	

** $P < 0.01$, * $P < 0.05$, [†] $P < 0.10$.

Table 6 Coefficient for the dependent variable: internal exploitation

	<i>Model 8a</i>		<i>Model 8b</i>		<i>Model 8c</i>		<i>Model 8d</i>	
	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>	<i>Beta</i>	<i>SE</i>
Constant	0.23	0.33	0.43 [†]	0.24	0.45 [†]	0.26	0.33 [†]	0.20
Employees	−0.03	0.03	−0.05 [†]	0.02	−0.04 [†]	0.02	−0.04 [†]	0.02
Net income	−0.02	0.02	−0.02	0.02	−0.02	0.02	−0.01	0.02
Specialization	0 [†]	0.00	0 [†]	0.00	0 [†]	0.00	0 [†]	0.00
Age	0 [†]	0.00	0 [†]	0.00	0	0.00	0 [†]	0.00
Solvency	0.01	0.12	0	0.13	−0.02	0.12	−0.02	0.12
Non-FC patents	0.01	0.03	0.01	0.03	0.01	0.03	0.01	0.03
Ext. Expr. (Function)			0.11*	0.05				
Ext. Expt. (Function)			−0.05	0.07				
Ext. Expr. (Structure)					0.07	0.07		
Ext. Expt. (Structure)					−0.01	0.07		
Ext. Expr. (Attribute)							0.05	0.06
Ext. Expt. (Attribute)							0.05	0.08
R ²	0.345		0.379		0.369		0.357	
F	3.570		3.830		3.450		3.210	
P <	0.000		0.000		0.000		0.000	

** $P < 0.01$, * $P < 0.05$, [†] $P < 0.10$

exploration, at least in the case of those exploitative alliances involving familiar partners. Also, in support of Hypothesis 1b, we found firms exploring through R&D alliances to specialize internally in refining their technology and improving their efficiency (i.e., internal exploitation). Finally, results of Hypotheses 2 and 3 prove the beneficial impact of pursuing such complementary strategies across firm boundaries, while showing the negative performance consequences of implementing learning mechanisms whose primary goals are redundant.

It is also interesting to discuss the circumstances under which our hypotheses have not been supported. First, the

results suggest that firms investing in internal exploration do not seek for external collaboration to further exploit their research outputs. A possible explanation has to be searched in the peculiarities of the empirical setting of our study. The search for innovative solutions is so risky and cost-intensive that those firms embarking in internal, non-collaborative exploration share a tendency to protect their research results maintaining them within the organizational boundaries (Bos, 1996; Schlecht, 2003). Moreover, firms cooperating with partners from different industries and with whom they are not familiar, then reveal to be not likely to specialize internally. In other words, it is the content of

Table 7 Performance consequences of ambidexterity

	Model 9a		Model 9b		Model 10a		Model 10b		Model 11a		Model 11b	
	Beta	SE	Beta	SE	Beta	SE	Beta	SE	Beta	SE	Beta	SE
Constant	-3.1**	0.59	-3.14**	0.59	-3.12**	0.59	-3.18**	0.59	-3.12**	0.59	-3.22**	0.59
Employees	0.14*	0.06	0.13 [†]	0.07	0.14*	0.06	0.14*	0.07	0.15*	0.06	0.14*	0.07
Net income	0.00	0.03	0.01	0.03	0.00	0.03	0.01	0.03	0.00	0.03	0.01	0.03
Specialization	0.03**	0.01	0.03**	0.01	0.03**	0.01	0.03**	0.01	0.03**	0.01	0.03**	0.01
Age	-0.01**	0.00	-0.01**	0.00	-0.01**	0.00	-0.01**	0.00	-0.01*	0.00	-0.01**	0.00
Solvency	0.25	0.32	0.29	0.31	0.24	0.32	0.26	0.31	0.25	0.31	0.31	0.30
Non-FC patents	0.53**	0.09	0.54**	0.09	0.54**	0.09	0.54**	0.09	0.53**	0.09	0.55**	0.09
Int. Expt.	0.46**	0.18	0.13	0.19	0.5**	0.19	0.25	0.21	0.53**	0.18	0.14	0.19
Int. Expt.	-0.21	0.14	-0.34*	0.17	-0.2	0.14	-0.28 [†]	0.15	-0.18	0.14	-0.32 [†]	0.17
Ext. Expt. (Function)	0.22**	0.05	0.23**	0.06								
Ext. Expt. (Function)	0.13**	0.04	0.15**	0.04								
Ext. Expt. (Structure)					0.23**	0.04	0.22**	0.05				
Ext. Expt. (Structure)					0.12**	0.05	0.16**	0.05				
Ext. Expt. (Attribute)									0.28**	0.05	0.30**	0.06
Ext. Expt. (Attribute)									0.03	0.03	0.03	0.04
Int. Expt. × Ext. Expt. (Function)			-0.13*	0.15								
Int. Expt. × Ext. Expt. (Function)			-0.14*	0.12								
Int. Expt. × Ext. Expt. (Function)			0.07**	0.11								
Int. Expt. × Ext. Expt. (Function)			-0.10	0.15								
Int. Expt. × Ext. Expt. (Structure)							-0.10	0.06				
Int. Expt. × Ext. Expt. (Structure)							-0.11	0.15				
Int. Expt. × Ext. Expt. (Structure)							0.06	0.13				
Int. Expt. × Ext. Expt. (Structure)							-0.14	0.10				
Int. Expt. × Ext. Expt. (Attribute)											0.16	0.12
Int. Expt. × Ext. Expt. (Attribute)											-0.23	0.16
Int. Expt. × Ext. Expt. (Attribute)											-0.05	0.17
Int. Expt. × Ext. Expt. (Attribute)											0.11	0.16
Chi square	247.480		1,100.060		293.170		1,151.590		244.230		1,510.340	
Log likelihood	-601.540		-598.830		-602.570		-600.000		-603.050		-599.520	
P <	0.000		0.000		0.000		0.000		0.000		0.000	

** $P < 0.01$, * $P < 0.05$, [†] $P < 0.10$.

the interorganizational agreement to matter the most in driving internal exploitation, due to the short history that still characterizes the FC industry (Mytelka, 2003). Similarly, testing the performance consequences of cross-boundary ambidexterity, firms exploring and exploiting externally through agreements both with partners operating in different industries and new partners do not benefit from improved innovation performance while balancing with internal exploitation and exploration activities. Therefore, internal exploration generates a positive impact on the innovation performance of the firm only if complemented with R&D-based external exploration.

Contribution to the literature

Providing a new perspective on how to balance technology-based explorative and exploitative learning activities, this study contributes to the ongoing discussion of the concept of ambidexterity in the literature on strategic management. Although the idea of simultaneously operating in contrasting domains is pervasive in most of the literature on learning and organizational change (Gibson and Birkinshaw, 2004), research has only recently begun to acknowledge the relevance of an interlevel focus to the simultaneous pursuit of intraorganizational and interorganizational learning strategies (Holmqvist, 2003; Holmqvist, 2004). In fact, most of the literature so far has addressed the

theoretical and empirical investigation of ambidexterity either within or between organizations, neglecting the possibility that intraorganizational learning might create a basis for interfirm knowledge generation and vice versa.

Our findings question the idea that firms can balance exploration and exploitation only within a given domain, thus accounting for potentially redundant mechanisms across domains. Accordingly, we advanced the concept of cross-boundary ambidexterity, here defined as the ability of a firm to take advantage of the opportunities to simultaneously balance exploration and exploitation across its boundaries.

Literature on ambidexterity agrees that exploration and exploitation are fundamentally different activities in terms of mindsets, structural arrangements, and managerial orientations (Tushman and O' Reilly, 1996; O' Reilly and Tushman, 2004; O'Reilly *et al.*, 2009). Advancing the idea that ambidexterity can be achieved across organizational boundaries, our study proposes an alternative perspective on ways to accommodate conflicting pressures involving exploration and exploitation (Gupta *et al.*, 2006). In this sense, organizational boundaries and specialization in technology-learning activities across them serve a role similar to that of the buffers between internal organizational units prescribed by the literatures to support firms' concurrent exploration and exploitation efforts (Benner and Tushman, 2003).

Although beneficial to firms' performance and their ability to change consistently with environmental dynamics (O'Reilly and Tushman, 2007), ambidexterity, for effective internal management, requires the development of complex, costly organizational structures allowing structural differentiation (Jansen *et al.*, 2009). It is likely that generally positive interactive effects on the development of intra-organizational ambidexterity might actually turn out to be negative, due to the difficulties of mediating between such contrasting learning logics (Knott, 2002; He and Wong, 2004). By juxtaposing intra- and interorganizational exploration-exploitation, our study provides a further hint on how firms can overcome trade-offs in resource allocation (Cheng and Kesner, 1997) and knowledge generation (Rosenkopf and Nerkar, 2001) across organizational boundaries, especially in the turbulent, risky contexts we selected for our analysis.

Practical relevance

Our results also have implications for managerial action. Earlier conceptual research has suggested that exploitation tends to be overemphasized, especially by large companies, to the detriment of more explorative orientation (Levinthal and March, 1993; Benner and Tushman, 2003; Uotila *et al.*, 2009). Although they provide support for this contention, showing the positive, significant impact of internal exploitation on a firm's propensity to enter into exploitative alliances, our results suggest a more immediate method to overcome potential sources of learning myopia. Accordingly, cross-boundary ambidexterity can represent a strategic option combining the benefits of investing in both proprietary research and interfirm collaborations.

To better clarify our results, empirical evidence is full of examples supporting such organizational learning behaviour. As in 2007, CFC Solutions GmbH, a wholly owned subsidiary of the Tognum Group, developed and manufactured stationary FCs for environmentally friendly production of electricity and heat. At CFC Solutions, they have always believed that next to, for example, wind power and solar energy, FCs will rank high among new and sustainable energy supply technologies. Based on the above strategic orientation, CFC Solutions has focused intense attention on the internal development of FC technologies. Although throughout the years the company has declared its openness to partnerships, as strategic and technological co-operations is considered the most effective way to improve application and system solutions or to optimize individual components in mutually beneficial partnerships, thus balancing the internal exploitation of proprietary technologies.

Similarly, as of 2007, Ballard Power Systems, founded in 1979 to research and develop high-energy lithium batteries, is recognized as a world leader in the design, development, and manufacture of zero-emission proton exchange membrane FCs. Over time and in order to successfully commercialize its FC technology, Ballard worked with industry-leading partners and customers who integrated Ballard's key components into end-use products to deliver the benefits of clean, efficient, and reliable power in a wide variety of potential applications. One of Ballard's major collaboration was the Vehicular Fuel Cell Alliance, signed

in 1997 with innovative vehicle manufacturer DaimlerChrysler (formerly Daimler-Benz AG) and Ford Motor Company, one of the world's largest automakers and a leader in advanced electric vehicle drive train technology. The benefits of this alliance have been demonstrated over time by increased development activities among Ballard's partners and customers and their plans to accelerate commercialization. DaimlerChrysler, Ford, and Honda have been able to introduce fleet demonstration vehicles and have already delivered some to customers. Although an extreme example, the case of Ballard proves that firms actively engaged in an emerging and innovative technological trajectory might decide to pursue refinement and exploitation both internally and externally, that is, seeking external complementarities while remaining internally focalized. The above considerations are strongly supported by the latest strategic decision at Ballard, which in November 2007 agreed to sell the company's automotive FC assets to Daimler and Ford. 'This transaction enabled Ballard to concentrate on growth in FC applications which provide clean energy solutions in commercial markets', said John Sheridan, Ballard's President and CEO. Therefore, building strong and stable relationships with specific partners has served to maximize the process of value creation associated with balancing exploration and exploitation through ambidexterity.

The study provides a concrete alternative strategy to meet the challenge of improving efficiency while simultaneously exploring new domains for action and innovation. Since resources are scarce by definition and exploitation of established technological paths is easier, less risky, and more profitable in the short run than exploration, firms could decide to specialize internally, continuing at the same time, at the cutting edge of the technological frontier, choosing the right partners and the right content for an alliance. In other words, according to the model developed in the study, firms' interorganizational relationships should be consistent not only at the level of technological ferment or of the amount of internally available resources, but also of the kind of learning trajectory that a firm pursues within its organizational boundaries.

In a partially related context, cross-boundary ambidexterity provides input into intellectual property management. There is agreement that firms underutilize their technology assets, in that most of the granted patents get stuck in the R&D laboratories (Rivette and Kline, 2000). This means that even through internal exploration, firms cannot derive all of the potential value of their R&D efforts because they do not allow for exploitation. On the contrary, however, by engaging in licensing and other commercial alliances, they could exploit patents as an important source of revenues and competitive advantage.

Limitations and future developments

Building on existing research on organizational learning trajectories, this study offers a new perspective on balancing exploration and exploitation. Given both the benefits of simultaneously pursuing efficiency and innovation and the related difficulties due to resource scarcity and contrasting organizational pressures exerted at different levels (i.e., individual, organizational, and interorganizational), firms

are presented with an alternative balancing option, that is, to specialize internally and externally, while achieving ambidexterity across their organizational boundaries.

As our analysis shows, cross-boundary ambidexterity is not only practiced by firms, but also benefits a firm's innovative performance. In this way we add to the debate over the need for combining contrasting learning strategies, providing further empirical corroboration of the advantages of ambidexterity.

Finally, from a methodological perspective, our study tests an alternative theoretical framework in an unusual empirical setting. The emerging FC industry is of particular significance to the study, given the complexity of its innovation search activities, and the absence of prior large-scale organizational research focused exclusively on it. Although our findings are likely to apply to other emerging industries, sharing the need for large-scale investments despite the high degree of uncertainty and risk involved, future research on how the framework developed in this study applies to other contexts would be helpful.

FCs are still in a fluid stage of development, with firms competing both on time-to-market and over the development of a dominant design. The specific characteristics of this industry could have also generated the unexpected results we found in this study, that is, the lack of balance between internal exploration and external exploitation. In a field in which research to develop commercially viable solutions is pervasive, alliances are considered conducive to risk reduction rather than exploitation of complementary assets, pursuing technological consolidation. As a consequence, future research is needed to examine the usefulness of our propositions both in more consolidated industries and in emerging industries where commercial results are relevant.

Specific traits that distinguish the FC industry from others also generate other limitations that could be overcome if the framework developed in this paper could be extended to other empirical contexts. First, due to data limitations, we could not investigate the impact of cross-boundary ambidexterity on measures of performance other than patenting activity. Future studies could test our hypotheses on different performance measures representative of different knowledge stages along the innovation process (e.g., citation-weighted patents, new product development, and market performance). Similarly, more mature industries, with a longer history, could allow examination of their own temporal adjustments that effectively benefited or detracted from their processes of exploration and exploitation, accounting for the impact of environmental and organizational drivers of change. Moreover, other types of organizational search could be analysed to verify the extent to which cross-boundary ambidexterity applies to domains other than technological search.

Our study aimed to test cross-boundary ambidexterity and its related performance consequences. Given the structure of our data, we did not test for potential drivers explaining firm tendency towards cross-boundary ambidexterity. This is surely a critical point. Previous literature has identified both organizational arrangements and environmental conditions as constraints or drivers to ambidexterity. We leave it to feature empirical and theoretical research to elaborate more on the organizational

and environmental factors fostering or hindering the development of cross-boundary ambidexterity.

Finally, field studies and qualitative analyses could shed more light on the dynamics of cross-boundary ambidexterity, with a focus on determinants at different levels of analysis, processes within and between organizations, and the related impacts on strategic orientation and organizational structure.

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Notes

- 1 IPC used for draw all fuel cell patents are H01M008/00, H01M008/02, H01M008/04, H01M008/06, H01M008/18, H01M008/22, H01M008/24, H01M008/08, H01M008/10, H01M008/12, H01M008/14, H01M004/86, H01M004/88, H01M004/90, H01M004/92, H01M004/94, H01M004/96, and H01M004/98. They are subset of 'fuel cell classification' (H01M008) and 'electrodes classification' (H01M004).
- 2 Due to less specified nature of USPC to this specific technology, the search selected patents having 'fuel cell' in their abstract or claim part given that they are subset of USPC 429, which is 'electrical current producing apparatus, product and process'.
- 3 Patents were collected at a consolidated level, that is, comprising both patents applied by the parent firms and all the consolidated (majority-owned) subsidiaries.
- 4 Given the breadth and number of variables that were used, the size of the panel was necessarily adjusted downwards based on the specific model being investigated.
- 5 We run additional analyses using a $t+1$ window for performance impact. Since the results were consistently similar, we decided to use the above-mentioned decay time of 2 years, also because it was more consistent with the current stage of development of the FC industry.
- 6 We did the same with the usual 5-year time period and results did not change significantly. We decided to maintain the 3-year threshold consistently with the pace of change that characterizes FC technology, still in a fluid stage of development.

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